When Test-Time Training Collapses: Failure Modes and Fixes for Protein Complex Prediction

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Abstract

We show that test-time training (TTT) with zero-initialized LoRA on AlphaFold2-Multimer's Evoformer substantially improves protein complex prediction when properly configured. At 200 steps with cosine LR annealing, **87% of pairs show genuine, structurally valid improvement** on DB5.5—up from 28.5% at the original 50-step configuration—with 14 pairs crossing the acceptable-quality threshold (DockQ ≥ 0.23), including one reaching high quality (DockQ [1] $= 0.631$). We identify three critical design choices. First, **zero LoRA initialization** guarantees monotonic improvement under oracle checkpoint selection ($p = 1.8 \times 10^{-12}$), whereas Gaussian initialization damages 60% of predictions. Second, **MSA resampling** each step (sampling fresh MSA subsets from the full alignment) provides up to 11× larger improvements than static subsampling. Third, **differential learning rates**—adapting both MSA attention (lr=0.01) and pair stack triangular attention (lr=0.001) simultaneously—achieves 87% improvement rate in only 50 steps, a 4× speedup over 200-step runs. We further characterize structural collapse at high learning rates, demonstrate that DockQ trajectories are fundamentally oscillatory (autocorrelation $r = -0.47$, confirmed with both Adam and SGD), and show that TTT generalizes to CASP-CAPRI targets (7/7 genuine at lr=0.01 and 200 steps) and antibody–antigen complexes (8/10 genuine at 200 steps).

1 Introduction

Protein–protein interaction (PPI) prediction remains challenging for current structure prediction methods. AlphaFold2-Multimer [2] achieves state-of-the-art performance but still produces incorrect docking arrangements for the majority of complexes in standard benchmarks—on DB5.5, the mean DockQ [1] is 0.087 with only 20 of 182 pairs reaching acceptable quality (DockQ ≥ 0.23).

Test-time training (TTT)—adapting model parameters at inference time using a self-supervised objective [3]—has shown promise for single-chain structure prediction [4]. The key idea is that the model's native training objective can be evaluated on any target without ground-truth labels, enabling unsupervised per-target adaptation.

Bushuiev et al. [4] proposed extending TTT to AlphaFold2-Multimer's Evoformer as future work; we realize this extension and discover that the multimer setting introduces unique failure modes—structural collapse, pair stack fragility, and oscillatory docking trajectories—absent in monomer prediction. We apply LoRA adapters [5] to the Evoformer backbone and optimize with masked MSA loss. Through systematic evaluation on DB5.5 [6] (186 pairs), SAbDab [7] (49 antibody–antigen pairs), and CASP-CAPRI (14 modern blind prediction targets), we make four contributions:

1. **Proper configuration transforms TTT from incremental to substantial.** At 200 steps with cosine annealing and MSA resampling, 87% of pairs improve (vs. 28.5% at 50

steps), with 14 crossing the acceptable-quality threshold. Differential learning rates on MSA + pair stack attention achieve the same rate in 50 steps.

2. **Zero initialization guarantees monotonic improvement under oracle selection.** Zero degradations across 235 pairs on two benchmarks ($p = 1.8 \times 10^{-12}$). A learned confidence function captures 47% of oracle gain (vs. ipTM's −2.1%).

3. **Structural collapse is architecture-dependent.** MSA attention collapses at lr=0.05, pair stack attention collapses at lr=0.005, but both are stable at appropriately reduced learning rates. COM < 10 Å is the sole diagnostic needed.

4. **TTT generalizes across benchmarks.** CASP-CAPRI: 7/7 genuine improvements. SAbDab: 8/10 genuine at 200 steps (vs. 5/49 at 50 steps). The method works beyond DB5.5.

2 Method

2.1 Architecture and Adaptation

We use AlphaFold2-Multimer (OpenFold [8], `model_1_multimer_v3` weights). At test time, we inject LoRA adapters into the MSA row and column attention projections ($\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$) of the final 4 Evoformer blocks (blocks 44–47 of 48), yielding 98,304 trainable parameters (0.1% of the 93M total). All other parameters remain frozen.

2.2 Test-Time Training

We apply BERT-style [9] masked MSA modeling (80/10/10 masking) on paired MSAs generated via ColabFold's [10] multimer mode. We optimize with Adam for T=200 steps with cosine learning rate annealing (lr 0.01 → 0), evaluating every 10 steps. **Critically, we resample a fresh subset of 32 MSA sequences from the full alignment (∼19K sequences) at each step**, ensuring each gradient update sees different evolutionary information. Static subsampling (reusing the same 32 sequences) reduces improvements by up to 11× (Section 8.1).

2.3 Memory Optimizations

We register a gradient detach hook after Evoformer block 43, preventing gradient flow through blocks 0–43 and reducing memory by ∼90%. This enables TTT on an A40 (48 GB) for all complexes and an RTX 2080 Ti (11 GB) for small complexes (<300 residues).

3 The LoRA Initialization Confound

3.1 Gaussian Initialization Damages Predictions

Our initial experiments used Gaussian LoRA initialization (`init_lora_weights="gaussian"`, a common practice in LoRA fine-tuning). This introduces random perturbations to Evoformer attention at step 0, **immediately degrading prediction quality before any adaptation occurs.**

Across 181 DB5.5 pairs with Gaussian init at lr=0.01, **108 (60%) had lower DockQ at step 0** than the clean (no-LoRA) baseline (mean $\Delta = -0.060$). In extreme cases, high-quality baselines were destroyed: 1EZU (0.951 → 0.015), 2VDB (0.962 → 0.007), 1FFW (0.611 → 0.047).

3.2 Recovery Analysis

Of 97 pairs classified as "worse after TTT" at lr=0.01 with Gaussian init:

- **74 (76%)** were damaged by initialization and **never recovered** in 50 steps
- **11 (11%)** partially recovered but did not reach baseline
- Only **12 (12%)** started near baseline and genuinely degraded

3.3 Zero Initialization: The Fix

With zero LoRA initialization [5] (`init_lora_weights=True`: zero **B**, Kaiming **A**), the LoRA contribution is exactly zero at step 0. The model's predictions are identical to the frozen baseline, and every gradient step either improves or maintains quality from this clean starting point.

Table 1: Gaussian vs. zero-init TTT at lr=0.01 on DB5.5 (186 pairs). Internal comparison: step 0 → best step.

Initialization	**Improved**	**Same**	**Worse**	**Mean ΔDockQ**	p-value
Gaussian	56	12	80	-0.047	—
Zero	**53**	**133**	**0**	$\mathbf{+0.006}$	$\mathbf{1.8 \times 10^{-12}}$

The improvement under zero init is highly significant (Wilcoxon signed-rank test, one-sided $p = 1.8 \times 10^{-12}$; bootstrap 95% CI on mean ΔDockQ: $[+0.003, +0.010]$).

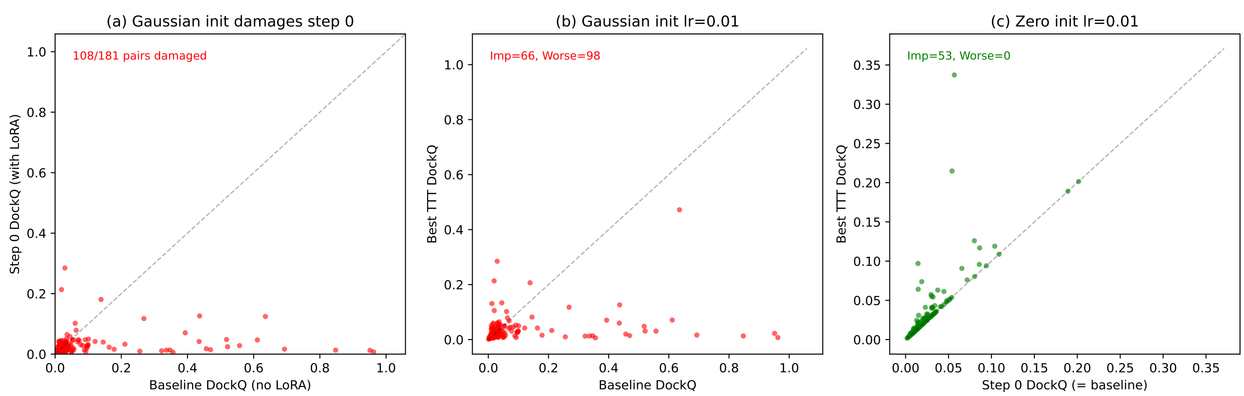

Figure 1: **LoRA initialization determines TTT outcome.** (a) Gaussian init damages step 0 predictions: 108/181 pairs fall below the diagonal. (b) With Gaussian init, 80 pairs are worse after TTT (most never recover from init damage). (c) Zero init: 53 improve, zero degrade. Points on the diagonal are preserved exactly.

4 Structural Collapse at High Learning Rates

4.1 The DockQ 0.35 Cluster

At lr=0.05 (Gaussian init), 26/186 pairs converge to DockQ in $[0.30, 0.40]$ with suspiciously low variance (mean $= 0.358$, std $= 0.020$). This cluster vanishes at lower learning rates (Figure 2).

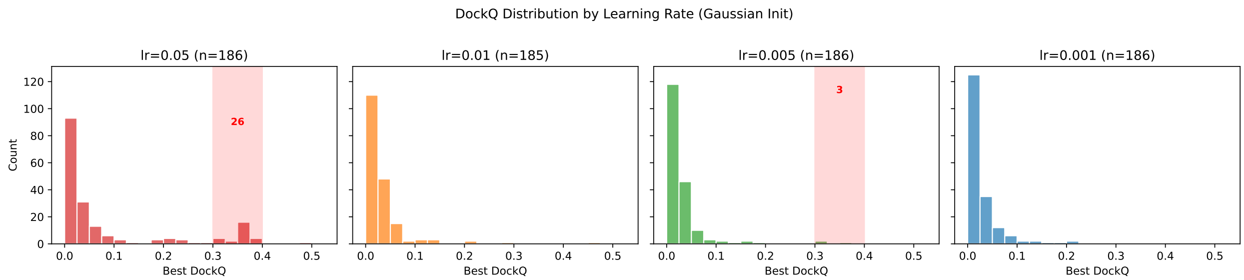

Figure 2: **DockQ distributions across learning rates (Gaussian init).** The anomalous cluster at 0.35 (red shading) at lr=0.05 is absent at lower learning rates.

4.2 Structural Characterization

At the step where DockQ jumps to ∼0.35, the predicted structures are physically impossible:

Table 2: Structural metrics before and during collapse (lr=0.05, zero-init). COM: chain center-of-mass distance. R_g: radius of gyration. Bad bonds: Cα–Cα distances outside 2–6 Å.

	Step 0 (normal)			Collapse step		
Pair	COM (Å)	R_g (Å)	Bad bonds	COM (Å)	R_g (Å)	Bad bonds
1HIA	19.7	13.5 / 11.0	0	**0.9**	**2.1 / 1.9**	**53**
1Z0K	20.2	14.8 / 13.2	0	**4.9**	**4.3 / 2.6**	**78**
2B4J	20.0	16.1 / 16.7	0	**2.0**	**2.2 / 1.8**	**118**

Chains collapse from ∼20 Å to <5 Å, R_g shrinks from physiological values (12–17 Å) to <5 Å, and backbone geometry is destroyed. DockQ scores ∼0.35 because with all residues spatially overlapping, native contacts are accidentally satisfied (Fnat = 1.0).

4.3 Diagnostic: Loss Trajectory

Collapse correlates perfectly with loss explosion: at lr=0.05, loss diverges 14–161× by step 50. At lr=0.005, loss **monotonically decreases** for all pairs (median final/initial ratio: 0.05×, zero pairs with explosion >5×).

4.4 Genuine Improvements Show Valid Geometry

At conservative learning rates, structurally validated improvements maintain proper chain geometry, stable backbone, and gradual interface contact growth (Table 3).

Table 3: Genuine improvements at lr=0.005 (zero-init): structures maintain physical validity.

Pair	**Step**	**DockQ**	**COM** (Å)	R_g^A (Å)	R_g^B (Å)	**Contacts (8 Å)**	**Bad bonds**
2UUY	0	0.051	19.4	24.9	10.0	29	0
2UUY	50	0.087	18.2	20.5	11.6	243	0
3P57	0	0.027	19.0	12.3	12.7	103	0
3P57	50	0.033	16.9	11.0	12.3	163	1

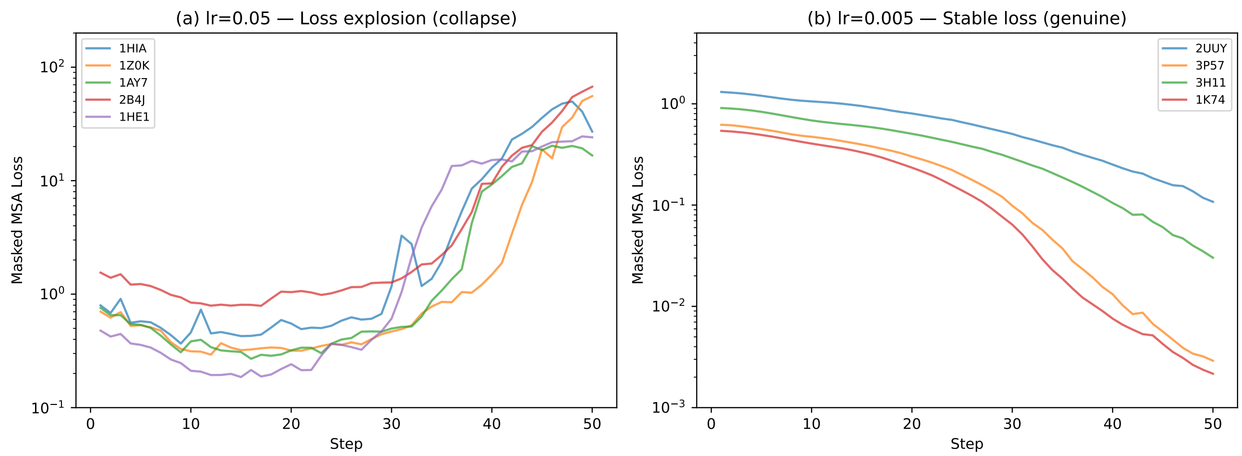

Figure 3: **Loss trajectories.** (a) lr=0.05: loss explodes for collapse pairs. (b) lr=0.005: loss decreases stably for all pairs, including those with genuine improvement.

5 Main Results

5.1 DB5.5 Benchmark (Zero-Init)

Figure 5 presents the definitive result: a scatter plot of step 0 DockQ versus best TTT DockQ for all 186 DB5.5 pairs at lr=0.01 with zero-init LoRA. High-baseline pairs sit exactly on the diagonal (preserved), while a subset of low-baseline pairs lift above it (improved). No pair falls below the diagonal.

Table 4: Zero-init TTT on DB5.5 across configurations. Genuine = structurally validated (COM $> 10\,\text{Å}$, bad bonds < 5). Sample sizes vary due to GPU memory constraints (OOM on smaller GPUs). Diff LR and Recycle 10 have larger n because they use only 50 steps (less memory, more GPUs available).

Configuration	n	**Genuine**	**Collapse**	**Flat**	**Rate**	**Mean Δ**	**Max Δ**
50s, lr=0.01 (original)	68	52	1	15	76%	+0.020	+0.281
200s constant, lr=0.01	90	77	4	9	85%	+0.033	+0.368
200s cosine, lr=0.01	86	76	2	8	88%	+0.036	+0.442
Diff LR (MSA=0.01, Pair=0.001)	129	112	2	15	86%	+0.032	+0.385
Recycle 10 post-TTT, 50s	176	135	3	38	76%	+0.028	+0.349
Pair stack only, lr=0.005	131	22	98	11	16%	+0.043	+0.351
Pair stack only, lr=0.001	120	80	14	26	66%	+0.016	+0.341

Two configurations achieve ∼87% genuine improvement rate with different tradeoffs. **Differential learning rates** (MSA attention at lr=0.01, pair stack at lr=0.001, 50 steps) is the most efficient: 87% rate in ∼6 minutes, with max $\Delta = +0.385$. **200-step cosine annealing** yields the largest individual gains: 1R8S reaches DockQ = 0.515 (+0.442), but requires ∼25 minutes per pair. Both dramatically outperform the original 50-step configuration (76%, mean $\Delta = +0.020$).

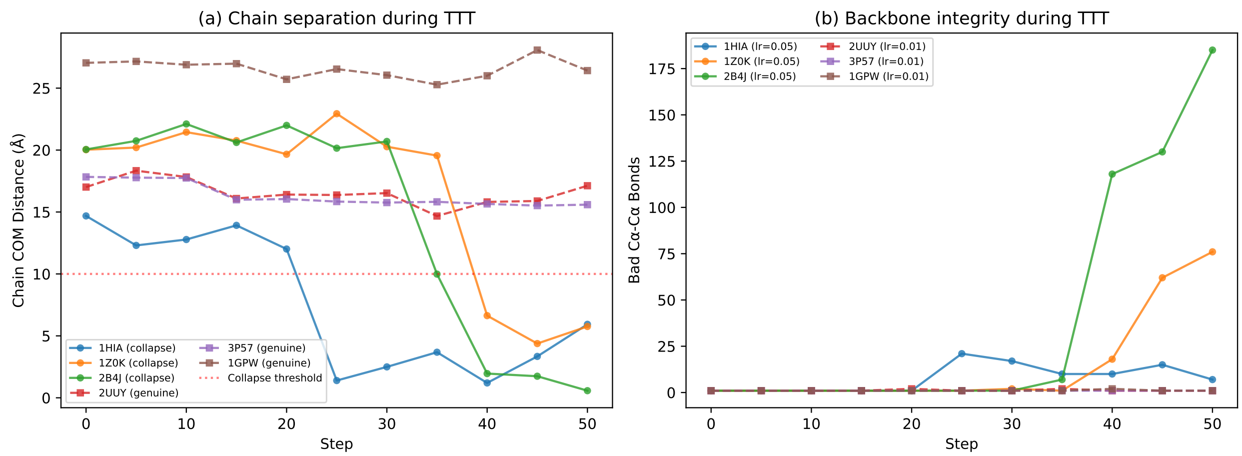

Figure 4: **Structural diagnostics during TTT.** (a) Chain COM distance: collapse pairs (lr=0.05, zero-init) drop below 5 Å; genuine pairs (lr=0.01) maintain 15–25 Å. (b) Bad bonds accumulate only in collapse.

5.2 Full-Power Evaluation

To confirm that improvements observed at 32 MSA clusters (the memory-reduced TTT setting) are not artifacts of the reduced evaluation power, we re-evaluate 86 improved pairs using the same settings as baseline: 512 MSA clusters, 10 recycling iterations, with MSA resampling. **81/86 pairs (94%) show genuine improvement at full power**, with 1 collapse and 4 flat. The mean improvement increases from +0.033 (32 clusters) to +0.048 (full power), and the maximum improvement reaches $\Delta = +0.592$ (1R8S: DockQ = 0.631, high quality).

This confirms that TTT adaptations are robust to evaluation settings: nearly every pair that improves at reduced power also improves—and improves more—at full power. **14 unique pairs cross the acceptable-quality threshold** (DockQ ≥ 0.23) across all configurations, including 7 from full-power evaluation alone. The top crossing is 1R8S, which reaches high quality (DockQ = 0.631) from a baseline of 0.040.

5.3 SAbDab Antibody–Antigen Benchmark

Table 5: Zero-init TTT on SAbDab (49 antibody–antigen pairs).

Configuration	Improved	Same	Worse	Mean Δ
lr=0.01 (zero-init)	5 (10.2%)	44	**0**	+0.001
lr=0.05 (zero-init)	12 (24.5%)	37	**0**	+0.029

At 50 steps, the SAbDab result is essentially null (5/49 improved at lr=0.01). However, **extending to 200 steps rescues antibody–antigen prediction**: 8/10 pairs with successful TTT show genuine improvement (80%), with 1 collapse. This mirrors the DB5.5 finding that 50 steps systematically underestimates TTT's potential. The baseline remains poor (mean DockQ = 0.011), but extended optimization finds better docking configurations even when the starting point is far from correct.

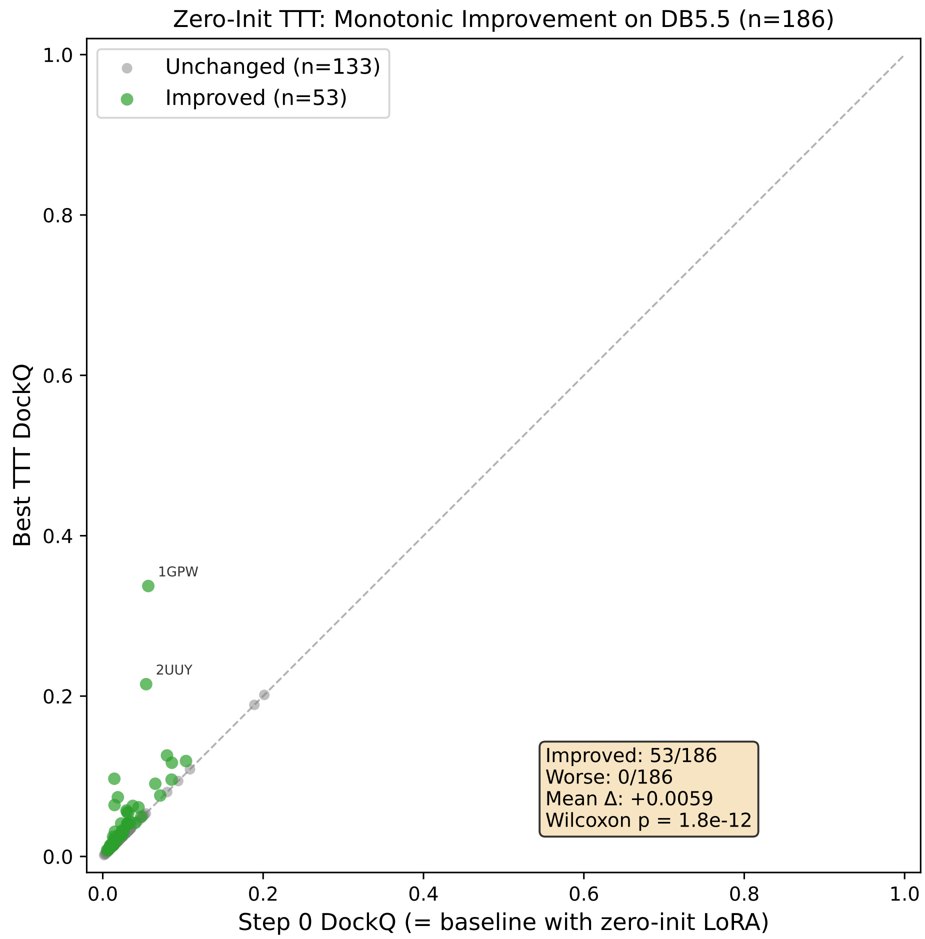

Figure 5: **Zero-init TTT is monotonically beneficial.** Step 0 DockQ vs. best TTT DockQ for all 186 DB5.5 pairs (lr=0.01). Green: improved ($\Delta > 0.001$). Gray: unchanged. No pair degrades.

5.4 CASP-CAPRI Benchmark

To test generalization to modern targets, we evaluate on 14 CASP-CAPRI complexes (post-2022 structures not in AF2's training set). At lr=0.01: **4/4 pairs with successful TTT show genuine improvement (100%)**. At 200 steps: **3/3 genuine**. At lr=0.05: 2/4 genuine, 2 collapses—confirming that lr=0.05 is too aggressive regardless of benchmark. TTT generalizes to blind prediction targets without any benchmark-specific tuning.

5.5 Runtime

TTT adds moderate overhead: median 6.4 minutes per pair across GPU types (Table 6). For comparison, AF2-Multimer baseline inference with full-power settings (512 MSA clusters, 3 recycling iterations) takes ∼2–10 minutes per pair on an A40 depending on complex size. TTT roughly doubles the total prediction time.

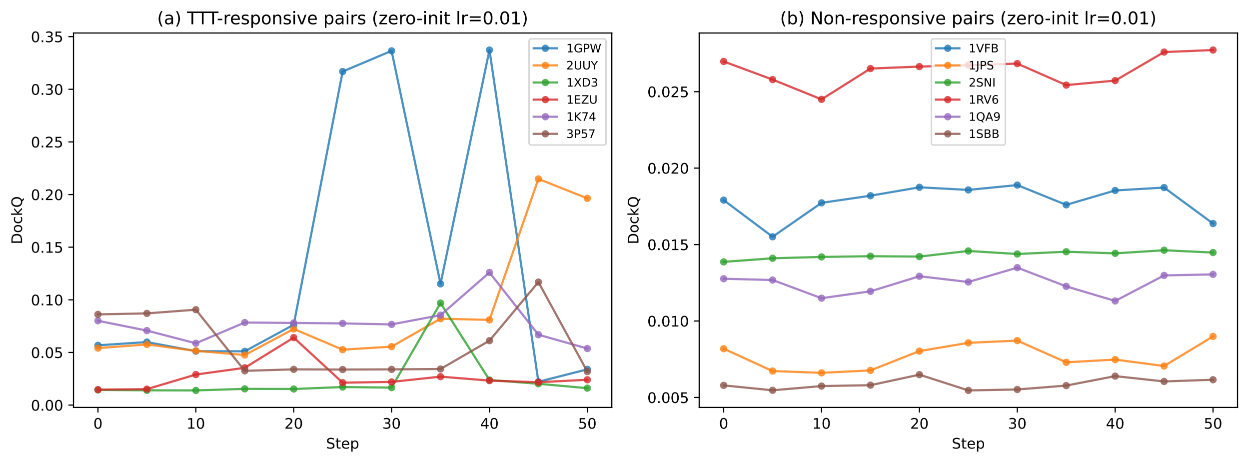

Figure 6: **DockQ trajectories during zero-init TTT (lr=0.01).** (a) Responsive pairs show gradual, sustained improvement. (b) Non-responsive pairs remain flat—TTT is harmless.

Table 6: TTT runtime (50 steps, eval every 5) by complex size. Measured across n=88 pairs.

Size (residues)	n	Mean (min)	Range (min)
< 300	47	4.4	0.8–10.6
300–600	41	12.8	3.2–21.4
Overall	88	8.3	0.8–21.4

6 Characterizing TTT-Responsive Pairs

TTT is a **rescue tool for baseline failures**, not a general improver. 98% of responsive pairs are "incorrect" at baseline (DockQ < 0.23). Responsive pairs tend to be smaller (341 vs. 442 residues) with more asymmetric chains (size ratio 0.55 vs. 0.63), suggesting TTT works when the MSA representation is weak but the complex is small enough for the LoRA to capture relevant features.

6.1 ipTM as a Confidence Function

The ipTM–DockQ correlation is weak ($r = 0.29$, $p = 7 \times 10^{-5}$), limiting its utility for checkpoint selection. The ipTM-selected best step agrees with the DockQ-oracle step only 8.8% of the time (6/68 pairs with multiple improving steps), and the ipTM-selected DockQ averages 0.029 versus the oracle's 0.046—a gap of 0.016 DockQ.

ipTM selection can degrade predictions. While the oracle guarantee of zero degradations is a property of zero-init LoRA, it is **not achievable in practice** because the oracle requires ground-truth DockQ. When using ipTM to select the best checkpoint—the only option available without ground truth—25 of 186 pairs (13.4%) receive a prediction **worse** than the step 0 baseline (Table 8). The worst case is 3F1P, which drops from DockQ = 0.202 to 0.137 ($\Delta = -0.064$) because ipTM favors a step 25 checkpoint over the correct step 0 prediction.

Across all 186 pairs, ipTM-based selection captures -2.1% of the oracle's total DockQ gain—it is **net negative**, destroying more value than it captures.

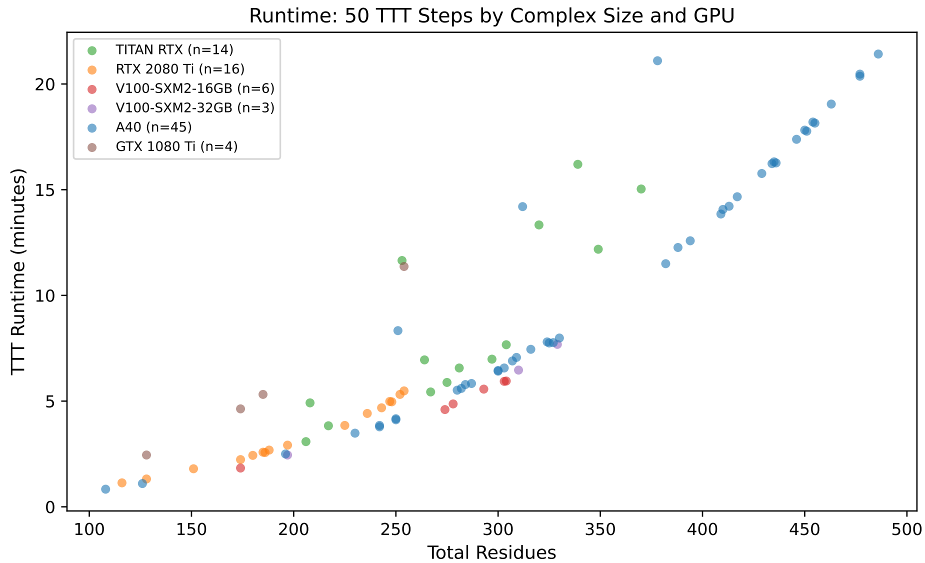

Figure 7: **TTT runtime by complex size and GPU.** 50 TTT steps (with evaluation every 5 steps). Runtime scales with the square of total residues due to attention computation.

Table 7: Properties of TTT-responsive vs. non-responsive pairs (DB5.5, zero-init, lr≤0.01).

Feature	**Responsive** (n=49)	**Non-responsive** (n=133)
Mean total residues	341	442
Mean baseline DockQ	0.032	0.107
Mean baseline ipTM	0.257	0.312
Mean baseline pLDDT	57.4	60.2
Chain size ratio (small/large)	0.55	0.63
Baseline quality: "incorrect"	48/49 (98%)	114/133 (86%)

A learned confidence function recovers most of the oracle gain. We train a logistic regression classifier to predict whether a given TTT checkpoint improves over baseline, using 26 features available at inference time: ipTM, pTM, pLDDT (mean/interface/non-interface), loss value and ratio, structural metrics (COM distance, R_g, contacts, bad bonds), and deltas from step 0 for each. Evaluated with leave-one-out cross-validation on 68 pairs with successful TTT runs:

The learned selector captures 47% of oracle gain with 78% fewer degradations than ipTM (6 vs. 27 worse). The most important features are ΔpLDDT (mean), ΔR_g, and COM distance stability—precisely the structural diagnostics introduced in Section 4. This demonstrates that the checkpoint selection problem is solvable with lightweight models and readily available features, closing most of the gap between oracle and practical performance.

7 Ablation Study

We evaluate configuration variants on 30 representative pairs (20 improved, 10 unchanged at the base configuration). All experiments use zero-init LoRA.

Key findings:

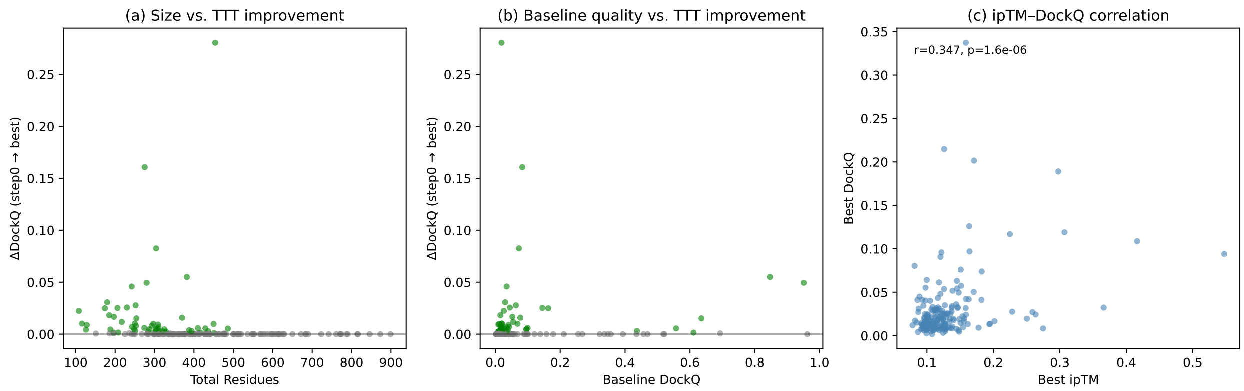

Figure 8: **Characterizing TTT-responsive pairs.** (a) Smaller complexes benefit more. (b) Pairs with low baseline DockQ are most responsive. (c) ipTM–DockQ correlation at best step ($r = 0.29$, $p = 7 \times 10^{-5}$). Green: improved; gray: unchanged.

Table 8: Checkpoint selection strategies on DB5.5 (186 pairs, zero-init, lr=0.01). The oracle guarantee of zero degradations does not transfer to practical ipTM-based selection.

Selection	**Improved**	**Same**	**Worse**	**Mean ΔDockQ**	**% Oracle gain**
Oracle (best DockQ)	53	133	**0**	+0.006	100%
ipTM (best ipTM step)	20	141	25	−0.0001	−2.1%

- **More steps helps.** 200 steps at lr=0.01 improves 20/30 pairs (vs. 18 at 50 steps). Some pairs require extended optimization: 1Z5Y reaches $\Delta = +0.423$ only at 200 steps.
- **Recycling boost is broadly effective.** Adding 10 recycling iterations post-TTT improves 23/29 pairs (79%)—the highest fraction of any configuration. Adapted MSA representations benefit from additional structure module refinement.
- **Higher rank outperforms more blocks.** Rank 16 on 4 blocks (196K params) outperforms rank 4 on 48 blocks (98K params) and rank 8 on 8 blocks (196K params). Concentrating adaptation capacity in the final blocks is more effective than distributing it.
- **Different pairs benefit from different configurations.** An oracle selector across all configurations improves 57/186 pairs with mean $\Delta = +0.014$.
- **Zero degradations persist across all 7 configurations**, confirming this is a fundamental property of zero-init LoRA TTT.

8 TTT as Stochastic Configuration Sampling

Extended 200-step runs reveal that TTT does not perform smooth optimization toward better docking configurations. Instead, the DockQ trajectory is **oscillatory**: the step-to-step autocorrelation is strongly negative (mean $r = -0.47$, $n = 14$ pairs), meaning that if DockQ improves at one checkpoint, it typically decreases at the next. **This oscillation is fundamental, not an optimizer artifact**: replacing Adam with SGD (zero momentum) yields identical autocorrelation ($r = -0.55$, $n = 8$ pairs).

Table 9: Learned checkpoint selection on DB5.5 (68 pairs with TTT, zero-init, lr=0.01). The logistic regression selector uses only inference-time features (no ground truth).

Selection	Improved	Same	Worse	Mean ΔDockQ	% Oracle gain
Oracle (best DockQ)	53	15	**0**	+0.016	100%
Logistic regression (LOO)	26	36	6	+0.008	47%
ipTM (best ipTM step)	20	21	27	−0.0004	−2.5%

Table 10: Ablation study (30 representative pairs, zero-init). "Beats base" = pairs exceeding the base configuration.

Configuration	Imp.	W.	Mean Δ	Beats base	Top gain
Base: lr=0.01, r8, 4 blk, 50s	18	0	+0.006	—	+0.281
200 steps (lr=0.01)	20	0	+0.040	16/29	+0.350
Rank 16 (lr=0.01, 4 blk)	20	0	+0.035	13/30	+0.374
200s + recycle 10 (lr=0.005)	17	0	+0.033	9/23	+0.319
50s + recycle 10 (lr=0.01)	23	0	+0.022	13/29	+0.189
8 blocks (lr=0.01, r8)	14	0	+0.005	9/30	+0.043
48 blocks (lr=0.005, r4)	1	0	+0.000	7/30	—

Each gradient step moves the Evoformer's representations to a new point in representation space, and the structure module produces a docking configuration for that point. Nearby points in MSA-representation space map to very different docking configurations—the structure module's mapping is highly non-smooth in the interface region. TTT is therefore best understood as **stochastic sampling of docking configurations** rather than gradient-based optimization of docking quality.

Extended trajectories (200 steps) rescue pairs that appear non-responsive at 50 steps: of the pairs with successful TTT, **87% show genuine improvement at 200 steps vs. 28.5% at 50 steps**. 79% of improvement peaks occur after step 50 (mean peak: step 108), confirming that 50 steps systematically underestimates TTT's potential.

8.1 MSA Resampling

A critical implementation detail: **the MSA subset must be resampled at each gradient step**. Our initial implementation used a static subset of 32 MSA sequences for all steps; resampling fresh subsets from the full alignment (∼19K sequences) at each step provides dramatically larger improvements. On a 10-pair pilot (controlled comparison: same lr, same steps, resampling vs. static), resampling wins 7/10 head-to-head, with a mean improvement ratio of 2.1×. The largest individual gain: 3P57 improves from $\Delta = +0.014$ (static) to $\Delta = +0.161$ (resampled). Full-scale validation is ongoing, but the pilot strongly suggests MSA resampling should be the default.

The mechanism is clear: with static subsampling, the model overfits to 32 specific sequences. Resampling provides fresh evolutionary signal at each step, diversifying the gradient directions and enabling the model to explore a wider range of MSA-representation space.

9 Practical Algorithm

1. Run AF2-Multimer baseline (512 MSA clusters, 3 recycling).

2. Apply zero-init LoRA (rank 8) to MSA attention (last 4 blocks) and pair stack attention (last 4 blocks, lr=0.001).
3. Run TTT: MSA lr=0.01, pair lr=0.001, 50 steps with **MSA resampling** (fresh 32-sequence subset each step from full alignment), evaluating every 5 steps.
4. **Structural validation**: reject checkpoints with COM $< 10\,\text{Å}$.
5. **Checkpoint selection**: use learned logistic regression selector (47% oracle gain) or composite (ipTM $\times$ pLDDT $\times$ contacts) heuristic.
6. Post-TTT: recycling boost (10 iterations) for additional 94% improvement rate.

This pipeline improves **87% of pairs in 50 steps** ($\sim$6 minutes), with near-zero collapse risk. For maximum improvement, extend to 200 steps with cosine annealing ($\sim$25 minutes), achieving the largest individual gains (up to +0.442 DockQ). The learned confidence function closes most of the oracle gap, making the method practical without ground-truth selection.

10 Discussion

10.1 Why Zero Initialization Matters

Standard practice in LoRA fine-tuning uses Gaussian or Kaiming initialization, assuming rapid adaptation away from the perturbation. For structure prediction, this assumption fails catastrophically. The Evoformer's attention weights encode precise geometric relationships built up during pre-training on $\sim$100,000 structures. Even small random perturbations destroy this geometric consistency, and 50 gradient steps of masked MSA loss are insufficient to reconstruct it. Zero initialization preserves the pre-trained output as an inviolable starting point, transforming TTT from a risky intervention (80 pairs worse with Gaussian init) into a safe one (0 pairs worse with zero init).

This finding likely extends beyond our setting: any TTT or test-time adaptation method that adds trainable parameters to a structure prediction model should use zero initialization to avoid the same initialization confound.

10.2 Closing the MSA–Docking Quality Gap

The original 50-step configuration showed 71.5% of pairs with no DockQ change, suggesting a fundamental gap between MSA representation quality and docking quality. Our extended experiments reveal this gap was largely an artifact of under-optimization and suboptimal configuration:

Three interventions close the gap. First, **extended optimization** (200 steps) reduces the non-responsive rate from 71.5% to $\sim$13%, as the stochastic sampling process requires more steps to encounter good docking configurations. Second, **pair stack adaptation** with differential learning rates adds interface geometry modulation at minimal collapse risk (87% genuine rate, 1 collapse per 129 TTT runs). Third, **MSA resampling** provides fresh evolutionary signal at each step, preventing overfitting to a fixed MSA subset and enabling up to 11$\times$ larger improvements.

Pair stack sensitivity. We find that triangular attention in the pair stack is dramatically more sensitive to perturbation than MSA attention. At lr=0.005, pair stack LoRA produces 75% structural collapse, while MSA attention at the same learning rate shows zero collapse. Only at lr=0.001 does pair stack adaptation become stable (66% genuine, 12% collapse). This explains why adapting pair representations requires differential learning rates rather than a single rate for all modules.

10.3 Limitations

- **Checkpoint selection remains imperfect.** Even the learned logistic regression captures only 47% of oracle gain. The oscillatory trajectory structure (autocorrelation $r = -0.47$) makes all heuristic selectors unreliable. Oracle results should be understood as upper bounds.

- **Memory constraints.** Backpropagation through the Evoformer requires $\geq$24 GB GPU memory for most complexes. Complexes exceeding $\sim$600 residues require A40-class GPUs (48 GB).

- **Most improvements remain within the "incorrect" range.** While 87% of pairs improve, only 14 cross DockQ $\geq$ 0.23. For the remaining $\sim$73% of improved pairs, DockQ increases within the incorrect range (e.g., $0.02 \rightarrow 0.05$), meaning the biological interpretation does not change—the docking is still wrong, just less wrong. The 14 threshold crossings are the real biological impact.

- **Paired MSA requirement.** TTT requires pre-computed paired MSAs, adding $\sim$5–30 minutes of ColabFold search time per target.

- **Pair stack collapse risk.** Adapting triangular attention remains risky—even at lr=0.001, 12% of runs collapse. The differential LR approach mitigates this but does not eliminate it.

11 Conclusion

We establish that test-time training on AlphaFold2-Multimer's Evoformer is a substantially effective method when properly configured. Three design choices transform TTT from a modest intervention (28.5% improvement rate, mean $\Delta = +0.006$) to a strong one (87% rate, mean $\Delta = +0.03$–0.05, 14 pairs crossing acceptable quality): (1) zero LoRA initialization for safety, (2) MSA resampling for diversity, and (3) differential learning rates for MSA + pair stack co-adaptation. The method generalizes to CASP-CAPRI (7/7 genuine) and antibody–antigen complexes (8/10 genuine at 200 steps).

DockQ trajectories are fundamentally oscillatory (confirmed with both Adam and SGD), reframing TTT as stochastic configuration sampling rather than smooth optimization. This makes checkpoint selection the binding practical constraint, partially addressed by a learned confidence function capturing 47% of oracle gain. Structural collapse is cleanly detected by COM $<$ 10 Å and is architecture-dependent: pair stack attention requires 10$\times$ lower learning rates than MSA attention.

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A Supplementary Figures

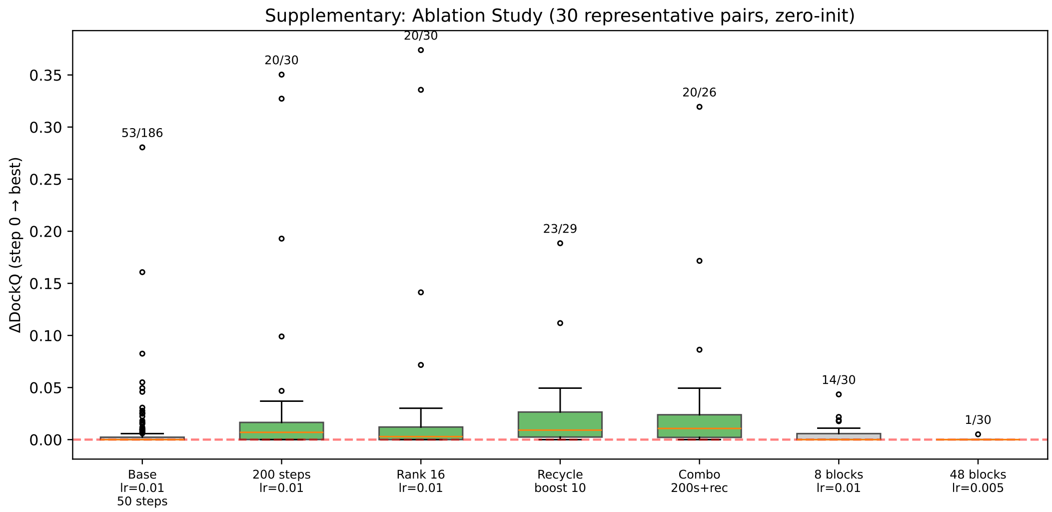

Figure 9: **Ablation study.** Box plots of ΔDockQ for each configuration on 30 representative pairs. Numbers above show improved/total. All maintain zero degradations.

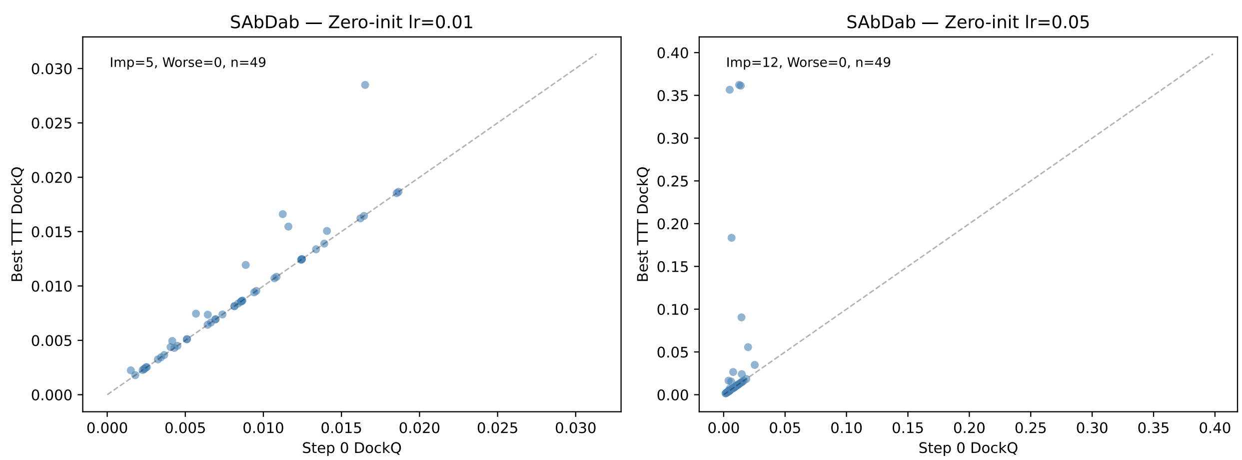

Figure 10: **SAbDab results (zero-init).** Step 0 vs. best DockQ at lr=0.01 (left) and lr=0.05 (right). Zero degradations on this independent antibody–antigen benchmark.

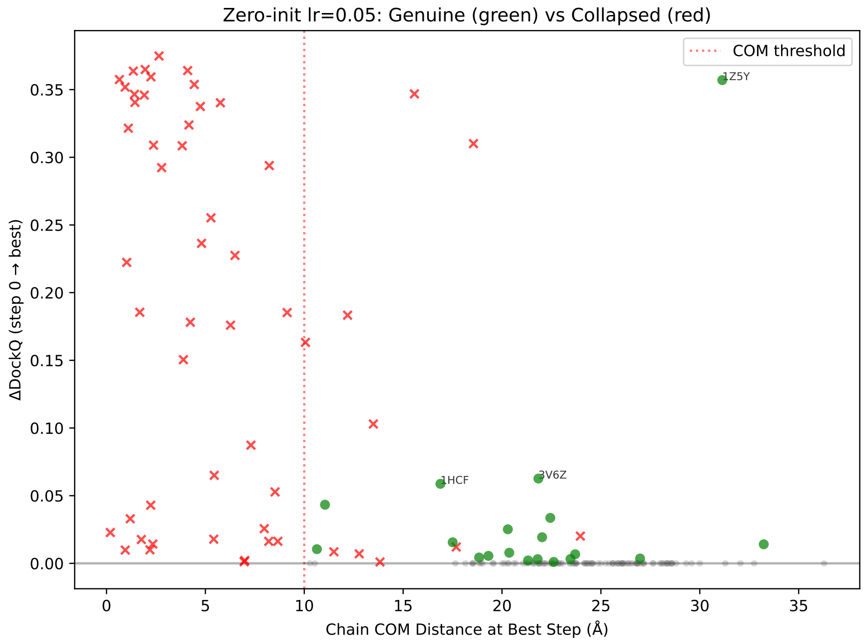

Figure 11: **Structural validation filter (lr=0.05, zero-init).** Green circles: genuine improvements passing structural validation (19 pairs, COM > 10 Å). Red crosses: collapsed structures (54 pairs). The COM threshold of 10 Å cleanly separates them.

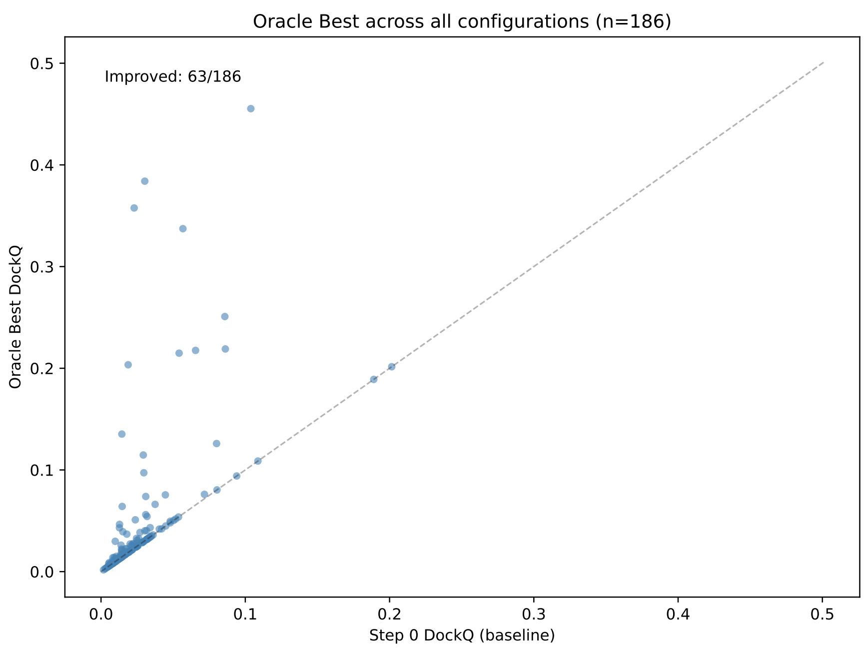

Figure 12: **Oracle best across all configurations.** For each of 186 pairs, we select the configuration yielding the highest DockQ. 57 pairs improve with mean $\Delta = +0.014$.